

Application User Guide

Sovereign On-Premise Agentic AI Workbench

How to Use the Application · How the Application Works

Core rule: AI may advise, retrieve, compare, summarize and reason — it never starts, stops, isolates, bypasses or controls plant equipment. **Humans decide.**

1. What This Application Is

The Sovereign On-Premise Agentic AI Workbench is a private, company-controlled industrial AI assistant for refinery environments (MRPL-style). It lets engineers, operators, maintenance teams, reviewers and administrators ask natural-language questions across SOPs, manuals, P&IDs, incident reports, maintenance history and sensor data — while keeping all company data and AI inference inside company infrastructure.

It is intentionally more than a RAG chatbot. Every answer is produced by a governed pipeline: deterministic safety checks first, then the cheapest trustworthy reasoning path (verified answer → cached approved context → hybrid retrieval → multi-document synthesis → full multi-agent reasoning), with citations, human approval where required, and a tamper-evident audit record of everything.

■ **This application is an advisory tool. It never executes plant actions and must not be treated as certified for live operational decisions without company approval.**

2. Getting Started

2.1 Access

- Open a browser on a company device connected to the LAN/VPN.
- Navigate to the workbench URL provided by your IT/AI platform administrator (e.g., `http://<company-server>:port`).
- No internet is required — the entire stack (frontend, API, databases, model) runs on the company server.

2.2 Login & First-Run Terms

- 1 Sign in with your company credentials. Access is role-based (RBAC): you only see what your role allows.
- 2 On first login, the **Terms & Conditions / Acceptable Use Policy** is shown. Scroll through the full text, tick the mandatory understanding checkboxes, and press **Accept**.
- 3 Your acceptance (user, timestamp, terms version, machine) is recorded in the tamper-evident audit chain.
- 4 You are taken to the Dashboard.

■ **Never share your credentials. All activity under your account is audit-logged and attributable to you.**

3. Using the Application — Step by Step

3.1 Dashboard

The Dashboard gives an overview: recent queries, pending approvals (if you are an approver), system/sovereignty status (local model active, hosted-inference count = 0), and quick links to console views.

3.2 Asking a Question (Query Console)

- 1 Open the **Query Console**.
- 2 Type your question in plain language, e.g.: *"P-204A vibration is elevated above baseline — what do the SOP, recent maintenance and sensor trends indicate?"*
- 3 Optionally narrow scope (equipment tag, document set, date range).
- 4 Press **Ask**. The deterministic gate checks the request first (instant).
- 5 Wait for the response. Fast/verified answers return in seconds; deep agentic analysis can take a few minutes — a progress indicator is shown.
- 6 Read the answer. Every claim is linked to cited evidence (document + revision, sensor window, maintenance record, P&ID/OCR region).

3.3 Reading the Answer

- **Citations:** each statement shows its source. Click a citation to inspect the exact chunk/record/revision used.
- **Observations vs. hypotheses:** sensor facts and historical records are labeled as observations; possible causes are labeled as hypotheses — never confuse the two.
- **Evidence sufficiency:** if evidence is missing, the system refuses or asks a clarifying question instead of guessing. This is intended, safe behavior.
- **OCR evidence:** shown with confidence scores and provenance; treat as indicative, not as proof of valve state or topology.

3.4 Human Approval Workflow (Action-Adjacent Advice)

- 1 When a recommendation could influence plant action (e.g., "inspect bearing, consider scheduling maintenance"), it is flagged **action-adjacent** and routed to an authorized approver (Operations Head / Shift-in-charge).
- 2 The approver opens the **Approvals** view, sees the full proposal with bound evidence (exact source revisions, hashes), and may **Approve**, **Reject**, or **Request changes**.
- 3 Only after approval is the advisory released back to the requester — clearly marked as approved, with approver identity and timestamp.
- 4 Nothing is sent to equipment at any point.

■ **The AI cannot approve its own outputs. If you are an approver, verify the cited evidence yourself before approving.**

3.5 Audit & Evidence Views

- **Audit view:** chronological, hash-chained record of queries, gate decisions, agent activity, approvals and releases. Entries cannot be altered or deleted by users.
- **Evidence manifests:** every governed output is bound to source identity, revision/hash and OCR provenance — click through from any approved output.
- **Sovereignty view:** confirms the active local model, local databases, and that hosted-inference dispatches are zero.

3.6 Ingesting & Managing Knowledge (Admins)

- Knowledge admins upload SOPs/manuals/drawings via the ingestion panel; the system parses (Docling/PyMuPDF), chunks, embeds and indexes them with revision metadata.
- Governed revisions: re-uploading a document creates a new revision; previously verified answers referencing the old revision automatically become stale.
- Verified Knowledge Registry: admins can mark reviewed answers as verified so repeated stable questions are served instantly without model reasoning.
- IT/AI admins manage users/roles, model runtime (Ollama), databases, backups and optional n8n side automation (notifications/schedules/exports only — never reasoning authority).

4. Do's and Don'ts

4.1 ■ Do

- Verify cited evidence before acting; use citations to open the source.
- Route consequential decisions through the approval workflow.
- Report missing or outdated documents to knowledge admins.
- Use the refusal/clarification responses as a signal to rephrase or narrow your question.
- Log recurring engineering lessons as shift notes/tacit knowledge for reuse.

4.2 ■ Don't

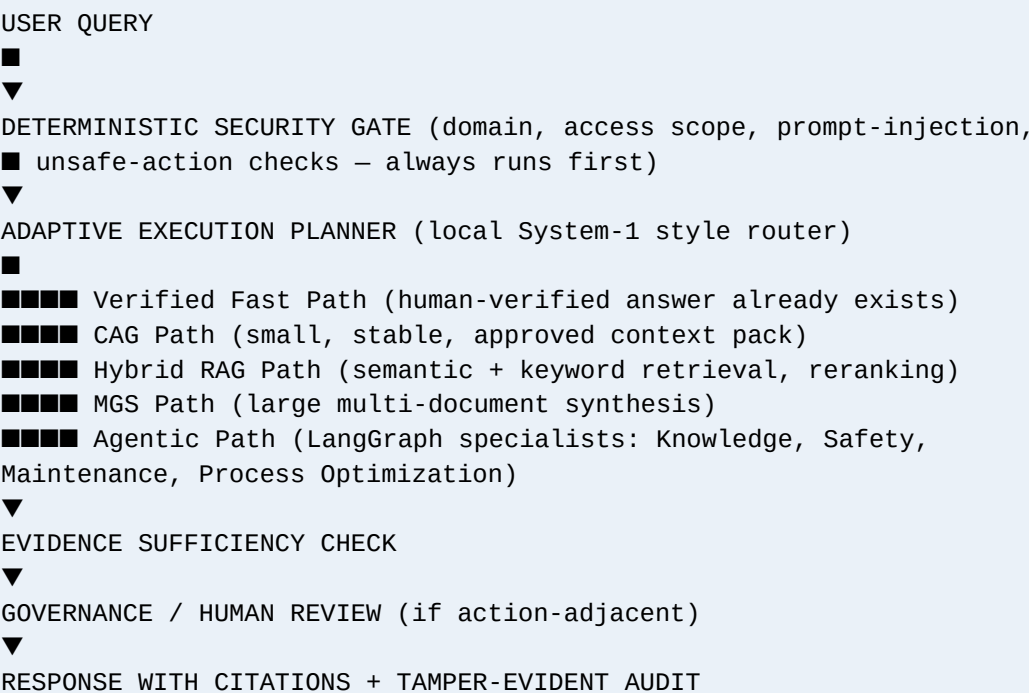
- Never attempt to make the system start, stop, isolate or control equipment — it has no such path, and attempts are logged.
- Don't try to bypass the safety gate, access scopes, or audit logging (attempts are recorded and reported).
- Don't treat OCR or image evidence as standalone proof of plant state.
- Don't paste company data into public AI services — this workbench exists so you don't need to.
- Don't share credentials or let unauthorized persons use your session.
- Don't treat AI output as a substitute for engineering judgment or mandatory procedures.

5. Who Uses It — Role Guide

- **Control-room / shift engineers:** ask operational questions, review SOP evidence, understand observations and limitations.
- **Maintenance / reliability engineers:** review maintenance history, equipment context, sensor anomalies and grounded hypotheses.
- **Safety / HSE teams:** retrieve procedures, correlate incidents, review safety routing and evidence-backed advisories.
- **Plant operations head / shift-in-charge:** act as final authority — approve/reject action-adjacent recommendations.
- **Knowledge / document administrators:** ingest, revise and govern company knowledge sources.
- **IT / AI platform administrators:** operate local models, PostgreSQL, Qdrant, audit, sovereignty controls and deployment.

6. How the Application Works

6.1 High-Level Flow



The router always picks the least-expensive path that can answer the request reliably. Simple repeated questions are answered instantly without invoking a large model; complex or safety-adjacent work escalates automatically.

6.2 The Safety Boundary (Cannot Be Overridden)

- The deterministic gate runs **before** any AI model and blocks out-of-domain, unsafe, injection or scope-invalid requests.
- Action-adjacent recommendations are routed to an authorized human approver; the AI cannot approve itself.
- Approval releases only a reviewed advisory — never an equipment command.
- OCR/image evidence always carries confidence and provenance and cannot silently become proof of plant state.
- Every query, decision and approval is written to a hash-chained, tamper-evident audit log.
- No hosted cloud AI (OpenAI/Gemini/Anthropic) is used; all inference runs on the local Qwen model via Ollama.

6.3 Data Storage (All Local)

Storage	What it holds
PostgreSQL	Users/roles, approvals, audit chain, equipment & structured records
Qdrant	Document embeddings & retrieval metadata for semantic search
Local files	Original PDFs, drawings, OCR artifacts, CSVs, evidence files

7. Known Limitations (Truthful)

- Validated on synthetic/public-style industrial data — not yet certified on live confidential plant data.
- No live SCADA/DCS write capability; the system is advisory only.
- OCR cannot independently establish as-built topology, isolation or valve state.
- Full agentic queries can take several minutes; fast/verified/CAG paths reduce latency for routine questions.
- Application-level sovereignty checks complement — but do not replace — company firewall/network policy.
- Advanced features (verified registry fast path, shift-handover intelligence, compliance agent, voice/multilingual interface, BI dashboards) are rolling out progressively.

■ If the system refuses to answer, it usually means evidence is insufficient or the request is out of scope — not a system failure.

8. Troubleshooting

Symptom	Likely cause / what to do
Page does not load	Check LAN/VPN connectivity; confirm the server stack is running (ask IT/AI admin).
Login fails	Verify credentials; contact admin for role/account issues.
Slow or no answer	Deep agentic queries can take minutes; check progress indicator. Retry with a narrower question to trigger a faster path.
Refusal / clarification prompt	Intended behavior — missing evidence or scope issue. Rephrase, narrow scope, or ask an admin to ingest the missing source.
Answer looks outdated	Check the cited revision; ask knowledge admin to re-ingest the current document version.
Sovereignty view shows hosted dispatches > 0	Escalate to IT/AI admin immediately — hosted inference must be zero.

9. Frequently Asked Questions

Q: Where does my data go?

A: Nowhere outside the company. Models, databases and files all run on company servers; hosted AI dispatch count is tracked and must remain zero.

Q: Can the AI operate equipment?

A: No. There is no execution path. It advises; authorized humans decide.

Q: What if the AI is wrong?

A: Always check citations. The system separates observations from hypotheses and refuses when evidence is insufficient. Report errors to admins so knowledge can be corrected.

Q: Why did my question take minutes?

A: Complex multi-evidence queries use full multi-agent reasoning. Routine questions are served via fast/verified paths once available.

Q: What gets recorded about me?

A: Queries, approvals and decisions are recorded in the tamper-evident audit chain for safety and governance, per the Terms & Conditions you accepted.

Q: Can I use it from home?

A: Only through the company-approved VPN, on an authorized device.

10. Glossary

RAG	Retrieval-Augmented Generation — answers grounded in retrieved company documents.
CAG	Cached approved context — small, stable, human-approved context packs for instant repeat answers.
MGS	Multi-generative-step synthesis — summarizes evidence groups then synthesizes with citations preserved.
HITL	Human-in-the-Loop — mandatory human review/approval for action-adjacent outputs.
Preflight gate	Deterministic pre-model checks: domain, access scope, injection, unsafe actions.
Evidence manifest	Binding of an output to exact source identity, revision/hash and OCR provenance.
Audit chain	Hash-chained, tamper-evident log of all queries, decisions and approvals.
Sovereignty	All inference and data remain inside company-controlled infrastructure.

Remember: Retrieve locally → Reason selectively → Verify with evidence → Approve with humans
→ Reuse only trusted knowledge.